

THANGAKUMAR DHANASEKARAN

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PROFESSIONAL SUMMARY

Software engineer with over a decade of experience building and shipping large-scale distributed systems across e-commerce, messaging, SaaS, and data platforms. Proven track record delivering highly reliable, scalable backend systems in Java, Python, and Node.js with deep expertise in distributed caching (Redis), RESTful API design, microservices architecture, and multi-region deployments. Passionate about engineering excellence at scale—comfortable navigating complex distributed systems problems, data-driven decision-making, and cross-functional collaboration. Direct experience in high-throughput e-commerce environments, including shipping, fulfillment, catalog, and cart services used by millions of buyers and sellers.

CORE COMPETENCIES

Large-Scale Distributed Systems • Java / Backend Engineering • RESTful & GraphQL API Design • Distributed Caching (Redis) • Microservices Architecture • Test Automation • E-Commerce Platforms • CI/CD & DevOps • Cross-Functional Collaboration • Data-Driven Engineering • Technical Mentorship

TECHNICAL SKILLS

Languages: Java, JavaScript/Node.js, Python, C#, C++, Groovy

APIs & Protocols: REST, GraphQL, HTTP/HTTPS, gRPC, Thrift — design, implementation, and versioning

Distributed Caching: Redis (caching layers, pub/sub, session management), RabbitMQ, Kafka

Databases & Storage: PostgreSQL, MongoDB, DynamoDB, Azure Cosmos DB, SQL Server, Trino, Apache Iceberg

Frameworks: Spring Boot, Express.js, React, ASP.NET Core, .NET Framework

Testing: Selenium, Playwright, JMeter, K6

Infrastructure & DevOps: Kubernetes, Docker, Jenkins, CI/CD pipelines, Rancher, Grafana, Kibana

Cloud: AWS (KMS, DynamoDB), Microsoft Azure (OpenAI, Cosmos DB)

Certifications: Azure Certified Developer Associate

PROFESSIONAL EXPERIENCE

ServiceNow | Senior Software Engineer

Jan 2024 – Present

- Owned the full development lifecycle—design, development, testing, and launch—for the ServiceNow Workflow Data Fabric (WDF) platform, enabling seamless cross-database connectivity via zero-copy connectors for high-availability data access.
- Designed and implemented scalable RESTful API backend services and React.js front-end applications, collaborating closely with product and design stakeholders to ship user-facing features on aggressive timelines.
- Built AI Agent-based automation tools and LLM-powered developer workflows (code analysis, test generation, intelligent documentation retrieval) that reduced manual overhead and accelerated team velocity.
- Championed a dev-owned quality culture by integrating TestNG, Rest Assured and Playwright automation frameworks into CI/CD pipelines—giving the engineering team full end-to-end delivery accountability from code commit to production release.
- Mentored engineers on distributed system design patterns and best practices, raising overall code quality through structured reviews and design discussions.

Samsung Electronics America | Staff Software Engineer

Jan 2023 – Dec 2023

- Led architecture and full-stack development of microservices for a large-scale e-commerce platform—Fulfillment, Shopping Cart, Orders, and Catalog services—delivering customer-facing features used by millions across the globe.

- Designed and maintained RESTful APIs (Express.js, ASP.NET Core) and React.js interfaces, owning code quality, documentation, and end-to-end test coverage from spec through production.
- Implemented distributed caching strategies using Redis to reduce database load and improve response latency for high-traffic product catalog and cart APIs—directly applicable to e-commerce shipping and inventory systems.
- Built and owned CI/CD pipelines (Jenkins, Docker, Kubernetes) that reduced deployment time by 50% and enabled reliable, zero-downtime releases.
- Drove a 40% increase in end-to-end test coverage and 30% reduction in query execution time through performance tuning, indexing strategies, and automation improvements.
- Mentored junior developers and conducted code reviews, contributing to team standards and best practices that improved engineering velocity.

LivePerson | Senior Software Developer

May 2021 – Jan 2023

- Led migration of monolithic Unified Messaging services to Kafka-backed Spring Boot microservices, improving fault isolation, independent scaling, and delivery speed for high-throughput message processing.
- Partnered with Verizon’s engineering team on integration testing and load validation for the Afiniti AI feature, coordinating cross-team technical design and resolving compatibility issues.
- Revamped Jenkins CI/CD deployment pipeline using Groovy parallel stages, cutting multi-region deployment time by 70% and enabling faster feature rollout.
- Upgraded Unified Messaging backend from JDK 8 to JDK 17, improving runtime performance and eliminating technical debt for long-term maintainability.

Samsung Electronics America | Engineer III

Sep 2018 – May 2021

- Built a scalable partner interaction RESTful microservice supporting PULL/PUSH inventory and price sync with external warehouse partners using Node.js, PostgreSQL, Redis, and RabbitMQ—handling high-throughput, distributed event streams at e-commerce scale.
- Developed a React/Express.js internal portal for end-to-end test automation, improving QA visibility for leadership by 50% and enabling faster issue triage.
- Strengthened platform security by migrating all credential management to AWS Key Management Service (KMS), reducing exposure risk for production secrets.

EARLIER CAREER

Senior Software Developer – Walgreens (via Cognizant) |

Jul 2010 – Sep 2018

Software Developer – Photon InfoTech | Software Engineer – Biztec Solutions LLC

8+ years delivering enterprise-grade web applications, RESTful APIs, and merchandising platforms across e-commerce and logistics. Built production monitoring systems with Kibana and comprehensive UI test automation frameworks using Selenium for high-traffic properties, including Walgreens.com, Drugstore.com, and Beauty.com. Technical proficiency across Java, C#, ASP.NET, MERN stack (MongoDB, Express.js, React, Node.js), SQL Server, and Oracle.

EDUCATION

Master of Science, Computer Science – University of Washington *(In Progress)*

Bachelor of Engineering, Electronics & Instrumentation Engineering – Anna University, India